When Regional Favoritism Fades

Nightlights and Pollution in Leader Birth Regions

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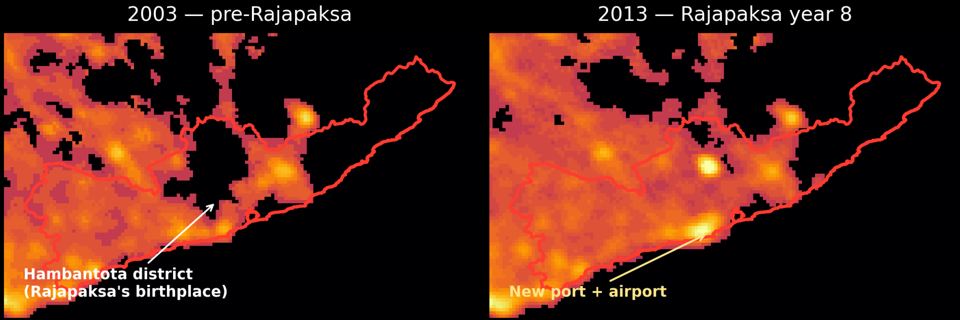
Mattala Rajapaksa International Airport

Built 2013 in rural Hambantota — President Rajapaksa's home district. $209M, financed by China. At opening it handled fewer than a dozen flights a week.

And from space



The airport, a $1.5B port, a cricket stadium, and a conference center all went up in Rajapaksa's home district. The 2013 nightlights show the new bright spot.

Why birth regions?

- Leaders direct resources to where they're **from**
- Hodler & Raschky (*QJE*, 2014) show this globally in satellite nightlights
- The effect is largest in Sub-Saharan Africa

This project

Research questions:

1. Does **birth-region favoritism** still show up in nightlights after 2005?
2. If not, did favoritism shift from *more* growth to *cleaner* growth? ("green favoritism")

Our contribution:

- **First systematic post-2005 test** of the Hodler–Raschky setup: effect replicates in 1992–2013 ($\beta = 0.013$, $p = 0.030$), but disappears after 2005 in every subsample we tested
- **Introduce green favoritism** as a novel composition-shift hypothesis — and rule it out with satellite NO_2 and $PM_{2.5}$

Data

Global ADM2 district-year panel combining:

- **PLAD**: leader identity, spell timing, subnational birthplace
- **Nightlights**: DMSP 1992–2013; harmonized DMSP–VIIRS 2005–2019
- **NO_2**: ACAG surface NO_2, 2005–2019
- **$PM_{2.5}$**: ACAG SatPM, 1998–2024
- **Geography**: GADM ADM2 boundaries

Empirical strategy: two-way fixed-effects DiD

$$\log(Y_{ict} + 0.01) = \beta \cdot \text{BirthRegionLeader}_{ict} + \alpha_i + \gamma_{ct} + \varepsilon_{ict}$$

$BirthRegionLeader_{ict} = 1$ if district i in country c is the birth region of the leader in office in year t, 0 otherwise.

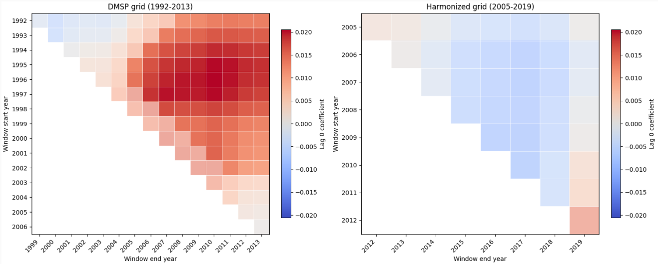
- ADM2 FE α_i, country-year FE γ_{ct}
- β compares the leader's birth district to other districts in the same country and year
- SEs clustered by leader spell
- Outcomes: log NL, intensity $=$ log NO_2 $-$ log NL

Result 1: replication holds, post-2005 collapses

Window	Coef.	SE	p	N
1992–2013	**0.0128**	0.0059	**0.030**	951,915
1995–2013	**0.0187**	0.0055	**0.001**	829,988
2000–2013	0.0110	0.0058	0.056	615,660
2005–2013	0.0017	0.0058	0.768	398,636
2005–2019	0.0004	0.0087	0.966	671,399

Long DMSP: $\sim$1.3–1.9% birth-region brightness premium. Post-2005: gone.

Result 2: attenuation is broad



Each cell is a separate regression.

- Positive estimates cluster in DMSP windows ending late 2000s
- Post-2005 windows: near zero everywhere

Not a single bad endpoint — a period-wide attenuation.

What could drive the attenuation?

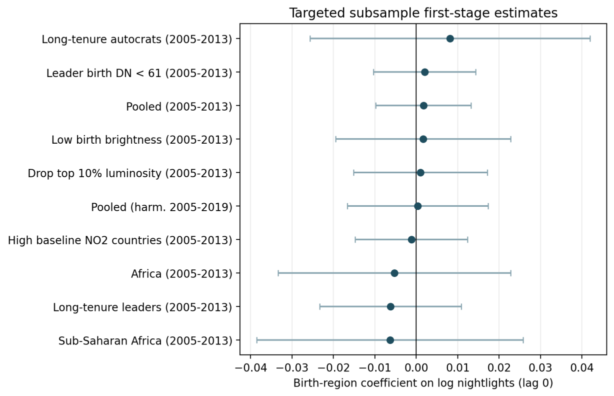
Four plausible stories:

1. **Democratization** — institutions check leaders' ability to favor home
2. **Africa-specific** — effect was largely an SSA story; maybe survives only there
3. **DMSP saturation** — birth districts too bright to register more growth
4. **Composition shift** — cleaner growth instead of more ("green favoritism")

Each story implies a test:

- Democratization $\Rightarrow$ autocrats only (Result 3)
- Africa story $\Rightarrow$ SSA subsample (Result 3)
- Saturation $\Rightarrow$ drop bright districts (Result 3)
- Green favoritism $\Rightarrow$ pollution outcomes (Result 4)

Result 3: no subsample rescues it



Searched where the effect should be strongest:

- Africa / SSA
- Long-tenure autocrats
- Low-brightness districts
- High-NO_2 countries

None passes coef. ≥ 0.005, $p < 0.10$.

Result 4: no green favoritism

If leaders shifted from more growth to cleaner growth, pollution should *fall* in birth regions.

Outcome (2005–2019, lag 0)	Coef.	SE	p	N
log Nightlights	-0.003	0.009	0.778	631k
log NO_2	$+0.031$	0.015	0.045	631k
NO_2 intensity	$+0.033$	0.018	0.063	631k
$PM_{2.5}$ intensity (lag 2)	$+0.018$	0.007	0.019	669k

Estimates go the **wrong way**.

Takeaways

1. The 1992–2013 result **still replicates**
2. After 2005 the effect collapses — and it doesn't come back when we slice the data
3. None of the four stories holds up:
    - Even autocrats don't show it $\Rightarrow$ not democratization
    - Even Africa doesn't show it post-2005 $\Rightarrow$ not just an Africa story
    - Even dim districts don't show it $\Rightarrow$ not saturation
    - Pollution isn't lower in birth regions $\Rightarrow$ not green favoritism
4. A textbook political-economy result looks much weaker once you push it past 2005

Broader implications & limitations

Social / policy implications:

- NGOs and aid monitors should watch for *green favoritism*, not just visible megaprojects — the *composition* of growth matters, not only the quantity
- Birth-region growth carries real pollution, so political patronage has an environmental-justice cost

Limitations:

- Reduced-form, not fully causal: identification rests on parallel within-country trends
- Birth region is one of many possible patronage targets (ethnic group, party base, capital city)
- Nightlights and satellite pollution are noisy proxies for true activity and exposure

Appendix: subsample activity estimates

Sample	Window	Coef.	SE	p
Africa	DMSP 1992–2013	0.0428	0.0163	0.009
Sub-Saharan Africa	DMSP 1992–2013	0.0442	0.0174	0.011
Non-SSA	DMSP 1992–2013	0.0053	0.0059	0.370
Africa	DMSP 2005–2013	−0.0052	0.0143	0.714
Sub-Saharan Africa	DMSP 2005–2013	−0.0063	0.0164	0.700
Long-tenure leaders	DMSP 2005–2013	−0.0062	0.0087	0.479
Long-tenure autocrats	DMSP 2005–2013	0.0081	0.0173	0.637
Drop top 10% luminosity	DMSP 2005–2013	0.0010	0.0082	0.899
Low birth brightness	DMSP 2005–2013	0.0016	0.0108	0.880
Africa	Harmonized 2005–2019	0.0262	0.0189	0.167
Sub-Saharan Africa	Harmonized 2005–2019	0.0315	0.0210	0.135